# Antidepressant Drugs
# Dr. Shanti Gurung

# Learning Objectives

- **Describe** neurochemical and neurotrophic theories of affect and their modulation by antidepressants.
- **Identify** major antidepressant classes and their cellular targets.
- **Apply** antidepressant therapy to common non-depressive disorders and conditions.
- **Compare** adverse effects and their mechanisms across antidepressant classes.
- **Recognize and manage** serotonin syndrome.

## ADRENALINE/EPINEPHRINE

**fight or flight**
Produced in stressful situations. Increases heart rate and blood flow, leading to physical boost and hightened awareness.

## GABA

**calming**
Calms firing nerves in the central nervous system. High levels improve focus, low levels cause anxiety. Also contributes to motor control and vision.

## NORADRENALINE/NOREPINEPHRINE

**concentration**
Affects attention and responding actions in the brain. Contract blood vessels, increasing blood flow.

## ACETYLCHOLINE

**learning**
Involved in thought, learning, and memory. Activates muscle action in the body. Also associated with attention and awakening.

## DOPAMINE

**pleasure**
Feelings of pleasure, also addiction, movement and motivation. People repeat behaviors that lead to dopamine release.

## GLUTAMATE

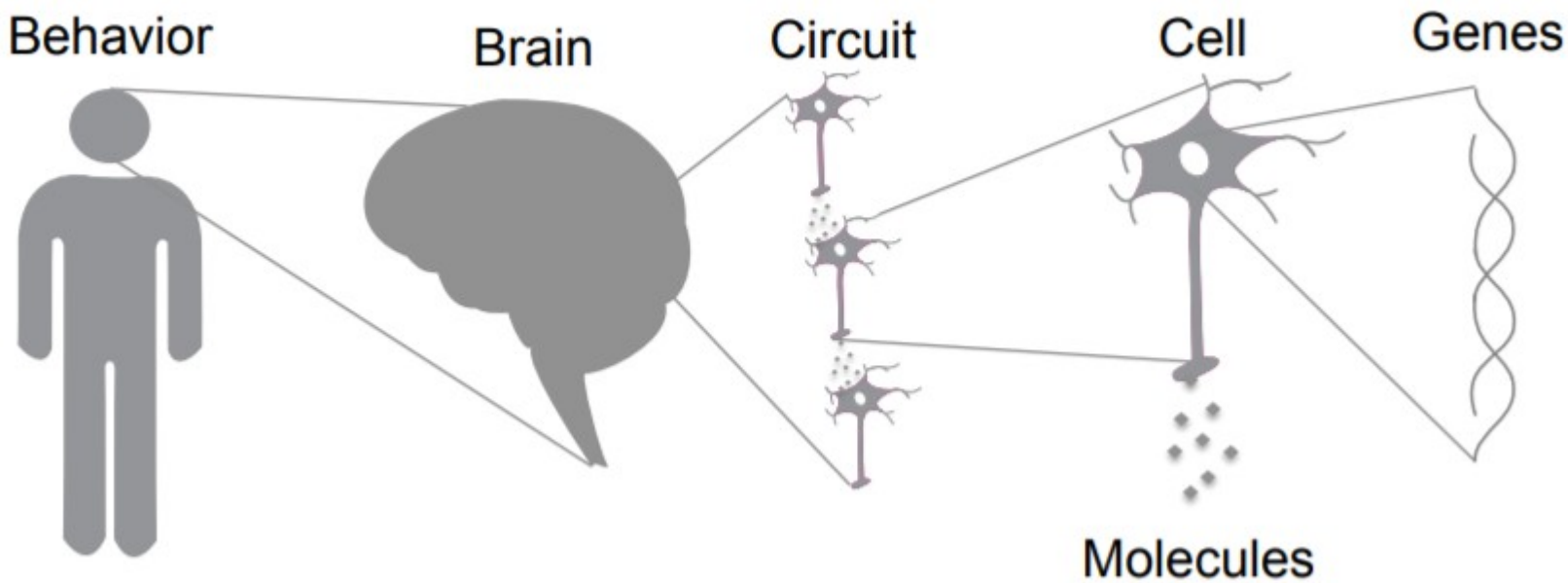
**memory**
Most common neurotransmitter. Involved in learning and memory, regulates development and creation of nerve contacts.

## SEROTONIN

**mood**
Contibutes to well-being and happiness. Helps sleep cycle and digestive system regulation. Affected by exercise.

## ENDORPHINS

**euporia**
Released during exercise, excitement and sex. Producing a sense of well-being and pain reduction.


Behavior
Brain
Circuit
Cell
Genes
Molecules

# Depression

Major depressive disorder (MDD) is characterized by depressed mood most of the time for at least 2 weeks and/or loss of interest or pleasure in most activities.

In addition, depression is characterized by disturbances in sleep and appetite as well as deficits in cognition and energy. Thoughts of guilt, worthlessness, and suicide are common.

# Clinical Presentation

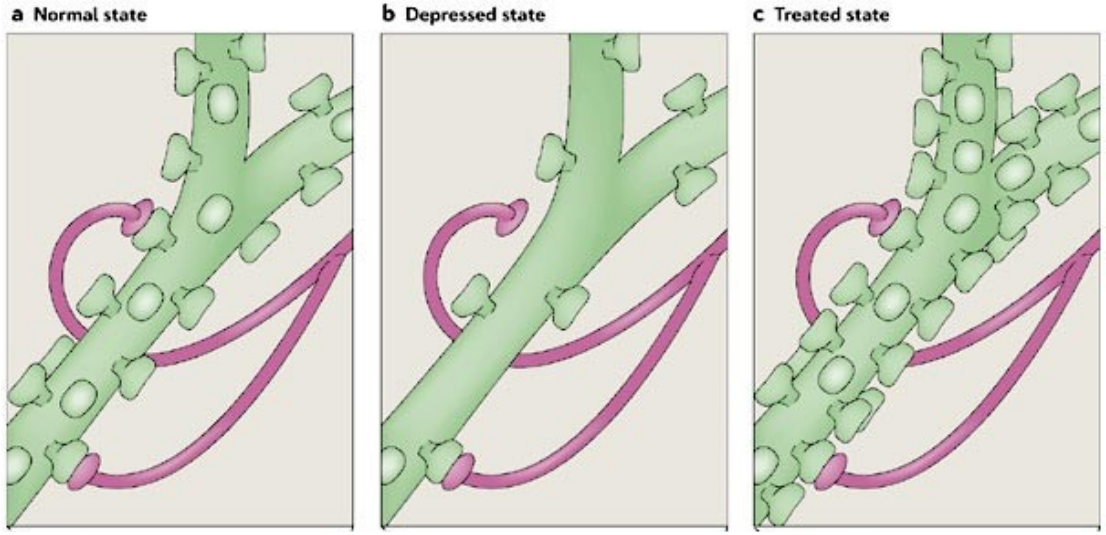
- Low energy level
- Sleep disturbances
- Lack of appetite
- Limited libido
- Inability to perform activities of daily living
- Overwhelming feelings of sadness, despair, hopelessness, and disorganization

# Pathophysiology of Depression

- Mono-Amine Hypothesis:

  Depression results from a deficiency in amount or function of cortical and limbic:

  - Serotonin
  - Norepinephrine
  - Dopamine



# Problems with Mono-Amine Hypothesis

- Antidepressant drugs (e.g., TCAs & SSRIs) have a prompt effect to increase synaptic levels of monoamines, yet the onset of their therapeutic effect is delayed for up to 3-4 weeks (or longer) in most patients

- The amine hypothesis does not adequately explain “how” or “why” a reduction monoamines occurs, or how it directly relates to symptoms of depression. Are changes in central monoamines a cause of depression, or an effect of depression?

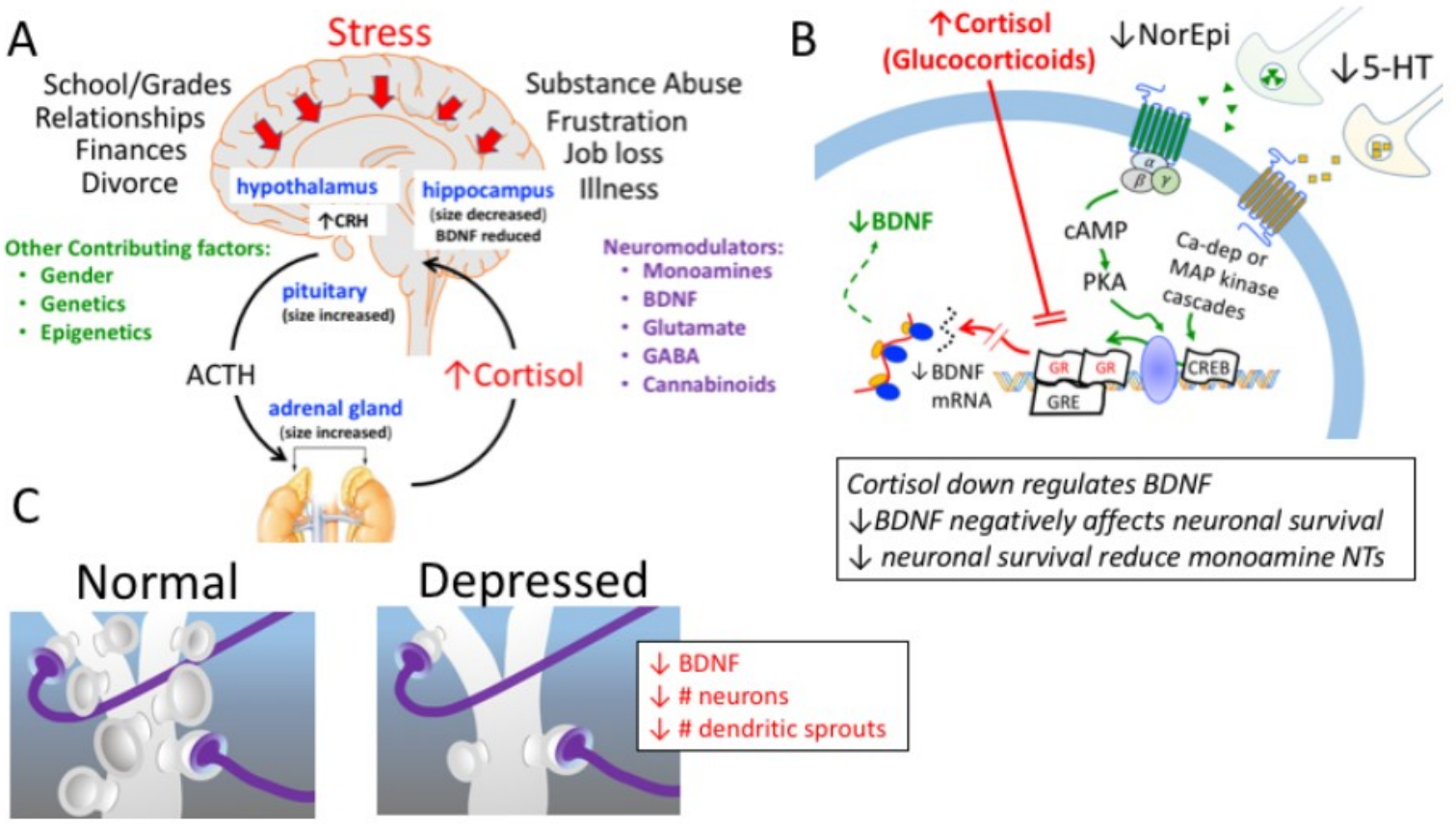
- The amine hypothesis does not account for newer observations of hippocampal atrophy and reductions in brain neurotrophic factors that are important for neuronal survival and plasticity

# Pathophysiology of Depression

- Neurotrophic Hypothesis:

  Reductions in BDNF result in neuronal atrophy and decreased synaptic density.

  Decreased BDNF levels and increased cortisol levels are believed to cause the reduced size of the hippocampus (a brain region involved in memory & mood control)




BDNF: Brain Derived Neurotrophic Factor

# Management of Depression

1. Psychotherapy

   Cognitive Behavioral Therapy (CBT) is one of the most common types of talk therapy used in the treatment for depression. CBT focuses on

   - **Education** (pinpointing the life problems that contribute to a patient's depression & which aspects of those problems they may be able to solve or improve)
   - **Improving self-esteem** (a patient's sense of self-regard, confidence & hope)
   - **Psychological (ego) functioning** (cognitive abilities, affect regulation, morals & ability to relate to others)
   - **Adaptive skills** (behaviors associated with effective function with friends, family & coworkers

2. Antidepressant Drug Therapy

# Antidepressant Drug Usage

- Antidepressant drugs were the most commonly prescribed medications in the USA (2005-07)
- Based upon meta-analysis of clinical trials, no single antidepressant or category of agents has been shown to be significantly superior in efficacy
- Their primary indication is for treatment of MDD. It has received FDA approvals for the treatment of panic disorder, generalized anxiety disorder (GAD), post-traumatic stress disorder (PTSD), and obsessive-compulsive disorder (OCD)
- Commonly used to treat pain disorders such as neuropathic pain and the pain associated with fibromyalgia

# Antidepressants: OLDER

## MAOIs

Phenelzine

Isocarboxazid

Tranylcypromine

Selegiline

## TCAs

Amitriptyline
Nortriptyline
Protriptyline

Clomipramine
Desipramine
Trimipramine
Imipramine

Doxepin
Amoxapine

MAOIs: Monoamine oxidase inhibitors
TCAs: Tricyclic antidepressants

# Antidepressants: NEWER

- SSRIs
  - Citalopram
  - Escitalopram
  - Fluoxetine
  - Paroxetine
  - Sertraline
- SNRIs
  - Venlafaxine
  - Desvenlafazine
  - Duloxetine
- SMS
  - Trazodone
  - Nefazodone
  - Vilazodone
  - Vortioxetine
- Tetracyclic/Unicyclic
  - Amoxipine
  - Bupropion
  - Maprotiline
  - Mirtazapine
  - Levomirtazapine

Serotonin modulators and stimulators (SMS)
(trazodone, nefazodone, vilazodone, vortioxetine)

Atypical anti-depressants
(bupropion, mirtazapine)

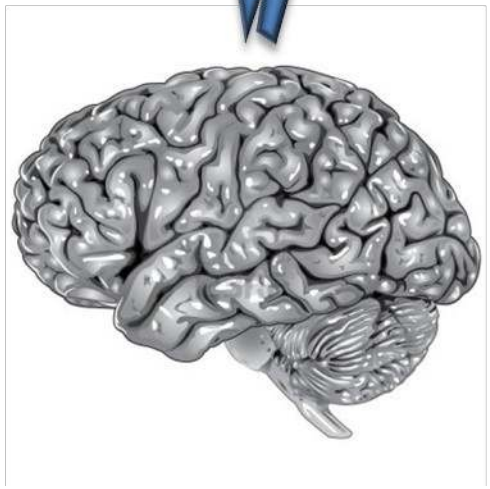
SNRIs
(venlafaxine)

SSRIs (citalopram)
NRIs (reboxetine)

TCAs
(imipramine)

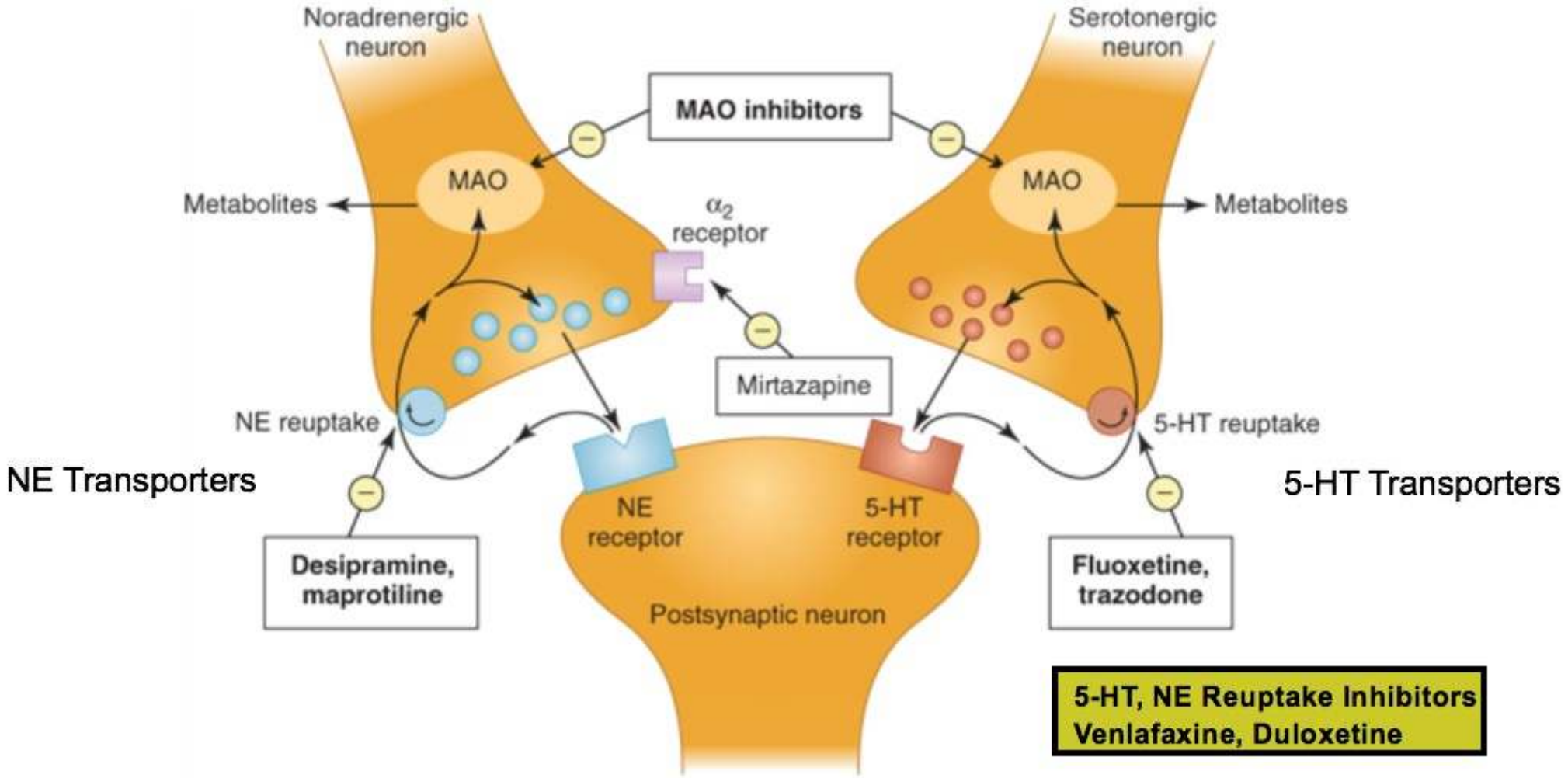
Reversible
MAO-A Inhibitors
(moclobemide)

Selective MAO-A
Inhibitors
(clorgyline)



Degree of
Adverse
Effects

# Antidepressants: MOA



# Clinical Issues

- The patient must understand the importance of continuing the drugs despite minimal initial response
- The lag time of 1-4 weeks between initiation of therapy and response must be emphasized
- Suicide precautions should be maintained until assessment indicates that suicidal ideation does not exist

# Selective Serotonin Reuptake Inhibitors (SSRIs)

Fluoxetine
Paroxetine

Citalopram
Escitalopram

Fluvoxamine
Sertraline

# Selective Serotonin Reuptake Inhibitors (SSRIs)

## Pharmacodynamics

SSRIs inhibit serotonin transporter (SERT)
Block only serotonin (not NE) reuptake- Elevated serotonin levels

## Pharmacokinetics

SSRIs is primarily metabolized by hepatic CYP P450 enzyme
The elimination half-life for most SSRIs is 20-30 hrs
Fluoxetine metabolism yields an active metabolite (**norfluoxetine**) (long half-life of 4-16 days)
SSRIs can produce **drug interactions** due to inhibition of P450-enzymes that metabolize other drugs

# Adverse Effects of SSRIs

Compared to TCA & MAOIs, SSRIs have "relatively benign" side effects due to their selectivity in affecting serotonin

Reported side effects are:

- Sexual dysfunction (17%)
- Drowsiness (17%)
- Weight gain (12%)
- Insomnia (11%)
- Anxiety (11%)
- Nausea (6%)

SSRIs can prolong the QT interval in a dose-dependent manner, although the increase is less than that associated with TCAs

Clinical studies indicate that **citalopram prolongs the $QT_c$** more than most other SSRIs

# Adverse Effects of SSRIs

SSRIs can produce drug interactions due to inhibition of CYP enzymes

The main P-450 isozymes inhibited by different SSRIs include:

Citalopram, Escitalopram: none
Fluoxetine: 2D6 (potent), 2C9, 2C19, 2B6,3A4
Fluvoxamine: 1A2 (potent), 2C19 (potent), 2B6, 2C9, 3A4
Paroxetine: 2D6 (potent), 2B6 (potent)
Sertraline: 2D6 (potent at high doses), 2B6, 2C9, 2C19, 3A4

SSRIs are contraindicated in patients who have received a MAOI in the previous 2 weeks due to risk of serotonin syndrome

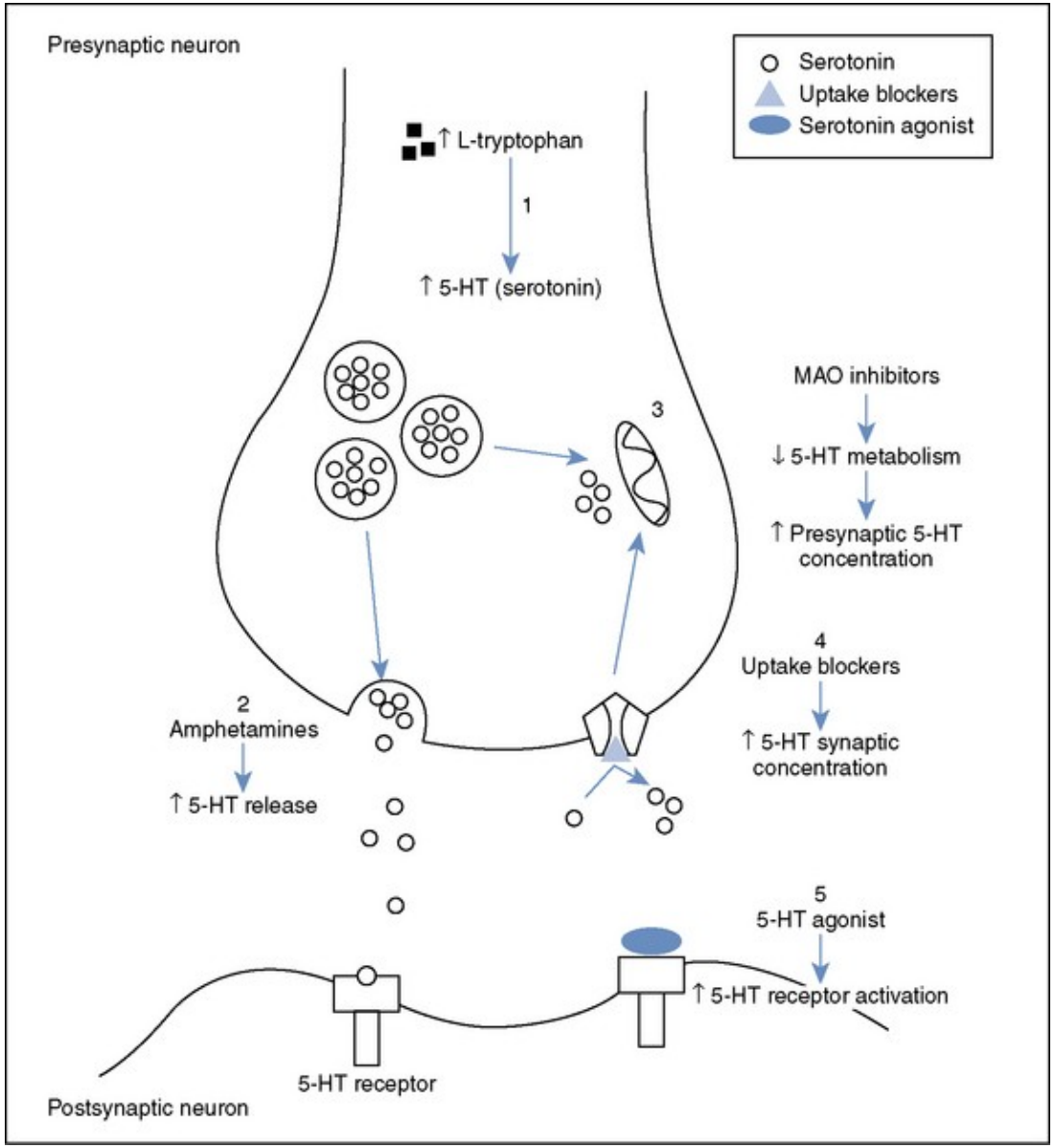
# Serotonin Syndrome

- It is a potentially life-threatening condition that is most commonly the result of patients taking multiple serotonergic drugs
- Drugs that can precipitate serotonin syndrome include :

| Amphetamines | Analgesics | Antidepressants | Antiemetics | Antimigraine | Miscellaneous |
|---|---|---|---|---|---|
| Dextroamphetamine | Cyclobenzaprine | MAO inhibitors | Metoclopramide | Carbamazepine | Buspirone |
| Methamphetamine | Fentanyl | SSRIs | Ondansetron | Ergot alkaloids | Cocaine |
| | Meperidine | SNRIs | | Triptans | Dextromethorphan |
| | Tramadol | Trazodone | | | Levodopa |
| | | St John's wort | | | Linezolid |
| | | TCAs | | | Lithium |
| | | | | | MDMA (Ecstasy) |

- Seven major serotonin (5-HT) subtypes have been identified that modulate behavior, thermoregulation, and mental status within the CNS, as well as GI motility, bronchoconstriction, vasoconstriction, and platelet aggregation peripherally

# Serotonin Syndrome



## Symptoms & Management in Serotonin Syndrome:

| Mild | Moderate | Life Threatening |
| --- | --- | --- |
| Mydriasis | **Altered Mental Status** (agitation, disorientation, excitement) | Delirium |
| Shivering | **Autonomic Hyperactivity** (rigidity, tachycardia, hyperthermia of >40°C) | Hypertension |
| Sweating | **Neuromuscular Abnormalities** (tremor, clonus, hyperreflexia) | Hyperthermia |
| Tachycardia (mild) | | Muscle rigidity |
| | | Tachycardia |

**Management stages:**

| | | |
| --- | --- | --- |
| Observe for at least 6 hrs<br>**Benzodiazepines** | Admit to hospital<br>Cardiac monitoring<br>**Cyproheptadine** | Intensive care unit<br>**Esmolol or nitroprusside**<br>**Cooling measures,**<br>Sedation, SkM paralysis, ventilation |

Meredith, a 24-year-old student who has been taking citalopram for depression arrives at the local hospital ED with a twisted ankle. Besides being in noticeable pain, she appears confused, with noticeable tremor & shivering, and is running a fever of 103°F. An overworked and inexperienced young physician establishes an i.v. and gives her meperidine to treat her shivering & pain. Soon thereafter Meredith becomes severely agitated and rips out her i.v. tubes. She is restrained, which sends her fever to lethal heights, ultimately contributing to her death.

What caused her initial symptoms?
What went wrong with her treatment?
How could her condition have been better managed?

What caused her initial symptoms?
What went wrong with her treatment?
How could her condition have been better managed?

In the case of Meredith:

- The patient's recent history was not adequately obtained prior to treatment, the clinical signs associated with serotonergic syndrome went unrecognized (improper clinical diagnosis)
- A pain medication (meperidine) which can increase serotonergic effects was given to treat shivering & pain, increasing the severity of her condition
- Restraining the patient (who was acting irrationally due to altered mental status) compounded her increased muscle rigidity, contributing to worsening of hyperthermia
- Agents to treat her agitation (benzodiazepines), serotonergic stimulation (cyproheptadine), and autonomic instability were not given in a timely manner

# Serotonin-Norepinephrine Reuptake Inhibitors

- Selective Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs)
  - Venlafaxine, Desvenlafaxine, Duloxetine

- Tricyclic Antidepressants (TCA)
  - Amitriptyline, Desipramine, Doxepin, Imipramine, Nortriptyline, Trimipramine, Clomipramine, Protriptyline

# Serotonin-Norepinephrine Reuptake Inhibitors

PD:

- Both TCAs and SNRIs bind both the serotonin (SERT) and the norepinephrine transporters (NET) and inhibit reuptake of serotonin and NE
- The SNRIs differ from the TCAs in that they lack the potent antihistamine, α adrenergic blocking, and antimuscarinic effects of the TCAs
- As a result, the SNRIs tend to be favored over the TCAs in the treatment of MDD and pain syndromes because of their better tolerability
- Among TCAs, clomipramine has more affinity to bind to SERT. This selectivity for the serotonin transporter contributes to clomipramine's known benefits in the treatment of OCD

# Side effects of SNRIs and TCAs

SNRIs-

- Serotonergic A/E (diarrhea, vomiting)
- Noradrenergic effects- ↑ BP and HR, and CNS activation, such as insomnia, anxiety, and agitation

TCAs-

- Alpha-1 Blockade: orthostatic hypotension, sexual dysfunction,
- Block cardiac K channels: QT prolongation (TdP)
- Histamine Blockade: weight gain, sedation
- Anti cholinergic: dry mouth, Blurred vision, constipation, urinary hesitancy
- Sexual Dysfunction
- Lowers seizure threshold: seizures may occur
- Overdoses TCA toxicity: 3 C's: Coma, Convulsions, Cardiac arrythmias (Rx- IV $NaHCO_3$)

# 5-HT2 Antagonists Trazodone, Nefazodone

Both Nefazodone and trazodone

- Block 5-$HT_2$ R potently (producing antidepressant, antianxiety & antipsychotic effects)
- α-1 receptor antagonist (causes orthostatic hypotension)

Trazodone

- Blocks 5-HT2 R (mainly due to its active metabolite m-cpp)
- Moderate - H1- receptor blockade (causes sedation)

A/E:

- Trazodone causes - mainly sedation and priapism

# Tetracyclic & Unicyclic Antidepressants

Bupropion – Weak reuptake inhibitor of NE & DA. No effect on 5-HT

- Indications:
  1. Smoking cessation effect- reduces the craving for nicotine
  2. Reversing SSRI- induced sexual dysfunction
- A/E: Lowers seizure threshold (contraindicated in epilepsy)

Mirtazapine –

- Effects:
  1. Antagonism of presynaptic α2 R- ↑ release of NE, 5-HT
  2. Antagonist of 5-HT2 and 5-HT3
  3. H1 antagonist (causing sedation and weight gain)
- A/E mirtazapine - sedation and weight gain

# Therapeutic Uses of Antidepressants

- Major depressive disorder
  - SSRIs, SNRIs, 5-HT antagonists, preferred over TCAs because of better tolerability
- Bipolar disorder (for depressive phase)
  - SSRI /TCAs in combination with Lithium (for maniac phase)
- Panic disorder
  - SSRIs, venlafaxine
- Obsessive compulsive disorder-
  - Clomipramine, SSRI (especially fluvoxamine)

# Therapeutic Uses of Antidepressants

- Enuresis (bedwetting)
  - Imipramine
- Attention deficit hyperactivity disorder(ADHD)
  - TCAs and Atomoxetine
- Nicotine dependence (to quit smoking)
  - Bupropion
- Neuropathic pain and fibromyalgia
  - Duloxetine, TCA
- GAD, panic attacks, social phobias, post-traumatic stress disorder, bulimia, and premenstrual dysphoric disorder
  - SSRI

A 27 yr old woman presents to a psychiatrist with a 20lb weight loss over the previous 2 months. She also complains of feeling of sadness, helplessness, lack of sleep for the past 1 month and had thoughts of suicide. The doctor asks her about her sleep patterns, ability to concentrate, energy level, mood interest level. Then the doctor explains to the patient that she has major depressive disorder, and he prescribes fluoxetine. Two days later the patient calls the doctor to indicate that the medicine is not working. He encourages her to continue taking the medicine after 2 more weeks the patient begins to feel better.

- What is the underlying neuropathological basis for the development of depressive symptoms in this patient?
- Why did the doctor encourage to continue taking the drug ?
- What is the mechanism by which fluoxetine alleviate these symptoms?

A. Inhibition of SERT
B. Inhibition of NET
C. Inhibition of 5HT2
D. Inhibition of MAO-B
E. Inhibition of MAO-A

BD is a 23-year-old woman who arrives at the University student health clinic for treatment of the flu. During the taking of her history, she admits that she has been feeling very sad & "worthless" for the past 3 weeks. She has also suffered from a loss of appetite, resulting in an unintentional loss of 10 lbs in the past few weeks. She has been experiencing "a lack of energy", and has been finding it difficult to sleep, or make decisions. She has also lost interest in her hobbies (jogging & tennis), and can't concentrate when studying for exams. She denies any substance abuse, which is supported by a negative tox screen of blood & urine samples. Based upon DSM-V criteria, BD is diagnosed with major depressive disorder, single episode with melancholic features. BD is given a prescription for escitalopram (Lexapro ®) 10 mg qd. How long should BD have to wait before a therapeutic response is most likely achieved?

? 1-2 days
? 1-2 weeks
? 3-4 weeks
? 3-4 months
? > 1 year

3-4 weeks

After 6 weeks, BD has a follow-up visit where she is asked about signs of improvement, as well as any side effects she has experienced. BD states that she has been sleeping better over the past 2 weeks. She recalls having mild transient bouts of nausea during the first week of treatment, but began taking her SSRI after breakfast, as instructed, and her nausea disappeared after several days. However she admits to having lost interest in having sexual relations with the fiancé, and recently has not been able to achieve orgasm (anorgasmia). You discuss the results of a clinical trial documenting improvement of sexual desire & reduced problems with achieving orgasm when a second drug is added. She elects to try augmentation with this drug. Which drug will you prescribe to produce this effect?

- ? bupropion (Wellbutrin ®)
- ? fluoxetine (Prozac ®)
- ? paroxetine (Paxil ®)
- ? sertraline (Zoloft ®)
- ? venlafaxine (Effexor ®)

bupropion (Wellbutrin ®)

DC is a 58 year old post-menopausal woman who is suffering from major depression. Her medical history indicates that she had an inadequate clinical benefit from previous trials of fluoxetine and imipramine. After discussion, you decide that she should start a trial with an SNRI after tapering her dose of fluoxetine. What drug will you prescribe?

- ? amitriptyline (Elavil ®)
- ? duloxetine (Cymbalta ®)
- ? escitalopram (Lexapro ®)
- ? paroxetine (Paxil ®)
- ? sertraline (Zoloft ®)

duloxetine (Cymbalta ®)

Doug is a 52 year-old man suffering from his first episode of major depression. Doug has a history of angina, and survived an MI three months ago. Which agent would be contraindicated for treating his depression?

? amitriptyline (Elavil ®)
? bupropion (Wellbutrin ®)
? duloxetine (Cymbalta ®)
? fluoxetine (Prozac ®)
? paroxetine (Paxil ®)

amitriptyline (Elavil ®)

Jimmy T is a 56-year-old professor with a history of hypertension who recently underwent a coronary artery bypass graft (CABD) procedure. Jimmy has a past history of cigarette smoking (1/2 pack per day for the past 20 years). During the recovery phase of his open heart surgery, Jimmy is diagnosed with major depression and seeks treatment. Which antidepressant may be able to treat both his mental disorder & help him decrease his craving for cigarettes?

? bupropion (Wellbutrin / Zyban ®)
? desipramine (Norpramin ®)
? fluoxetine (Prozac ®)
? nefaxodone (generic)
? sertraline (Zoloft ®)

bupropion (Wellbutrin / Zyban ®)

# THANK YOU